SPECIFICATION
FOR
AMOLED Module

MODULE No:	KD050HDOIN116-C000-a
CUSTOMER:	

STARTEK	INITIAL	DATE
PREPARED BY		
CHECKED BY		
APPROVED BY		

CUSTOMER	INITIAL	DATE
APPROVED BY		

Revision History

Date	Rev. No.	Page	Summary
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Contents

1. General Specifications 4
2. Block Diagram 5
3. Outline dimension 6
4. Input/output Terminals 7
5. AMOLED Optical Characteristics 9
5.1 Optical specification 9
6. Electrical Characteristics 12
6.1 Absolute Maximum Rating 12
6.2 DC Electrical Characteristics 12
7. AC Characteristics(MIPI) 13
7.1 HS Data Transmission Burst 13
7.3 Turnaround Procedure 14
7.4 Timing Parameters 15
7.5 Timing requirements for RESETB 16
8. Recommended Operating Sequence 18
8.1 Power On Sequence 18
8.2 Power off Sequence 19
8.3 Brightness control 19
9. AMOLED Module Out-Going Quality Level 20
9.1 VISUAL & FUNCTION INSPECTION STANDARD 20
9.1.1 Inspection conditions 20
9.1.2 Definition 20
9.1.3 Sampling Plan 21
9.1.4 Criteria (Visual) 22
10. Reliability Test Result 26
11. Cautions and Handling Precautions 27
11.1 Handling and Operating the Module 27
11.2 Storage and Transportation. 27
12. Packing 28

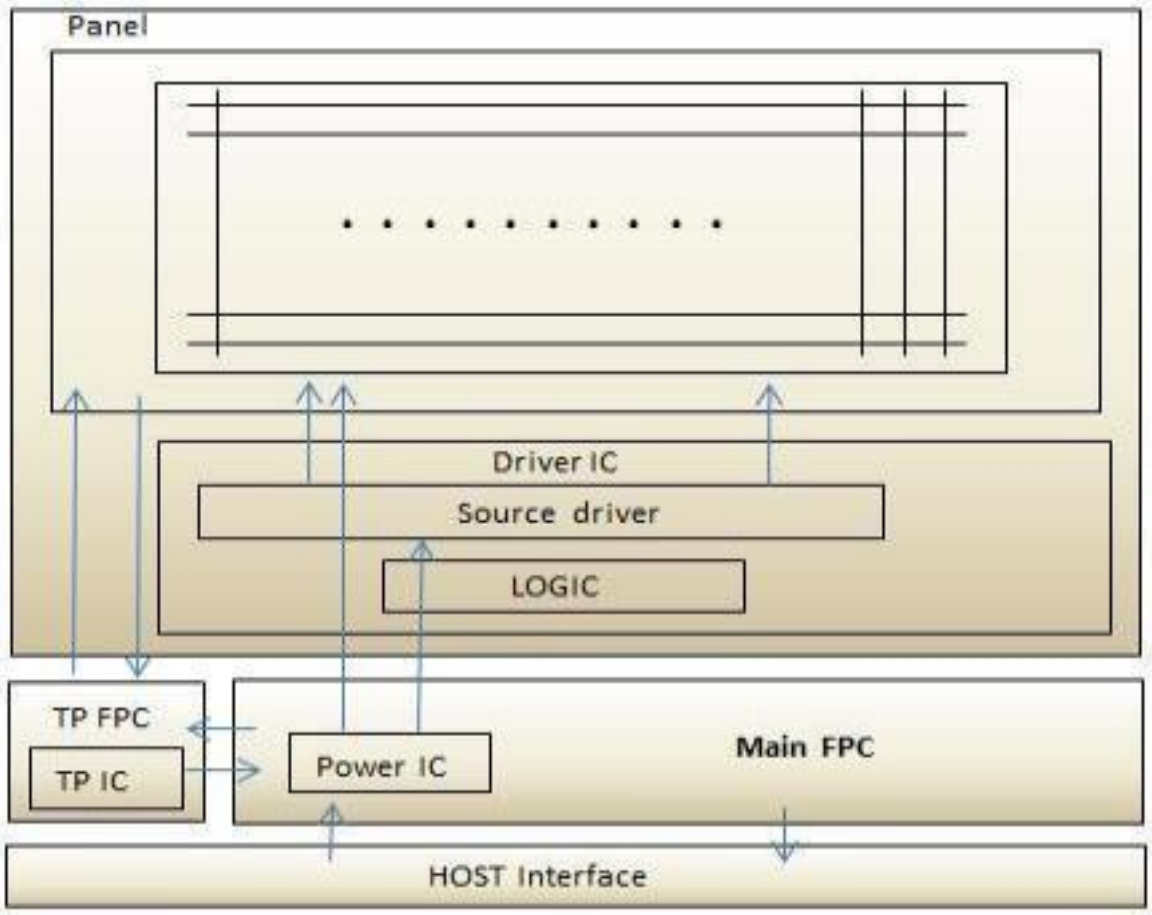
* Description

4.97” is a color active matrix AMOLED module using Low Temperature Poly-silicon Thin Film Transistors as active switching devices. The resolution of a 4.97” AMOLED contains 720x1280 pixels,and can display up to 16.7M colors.

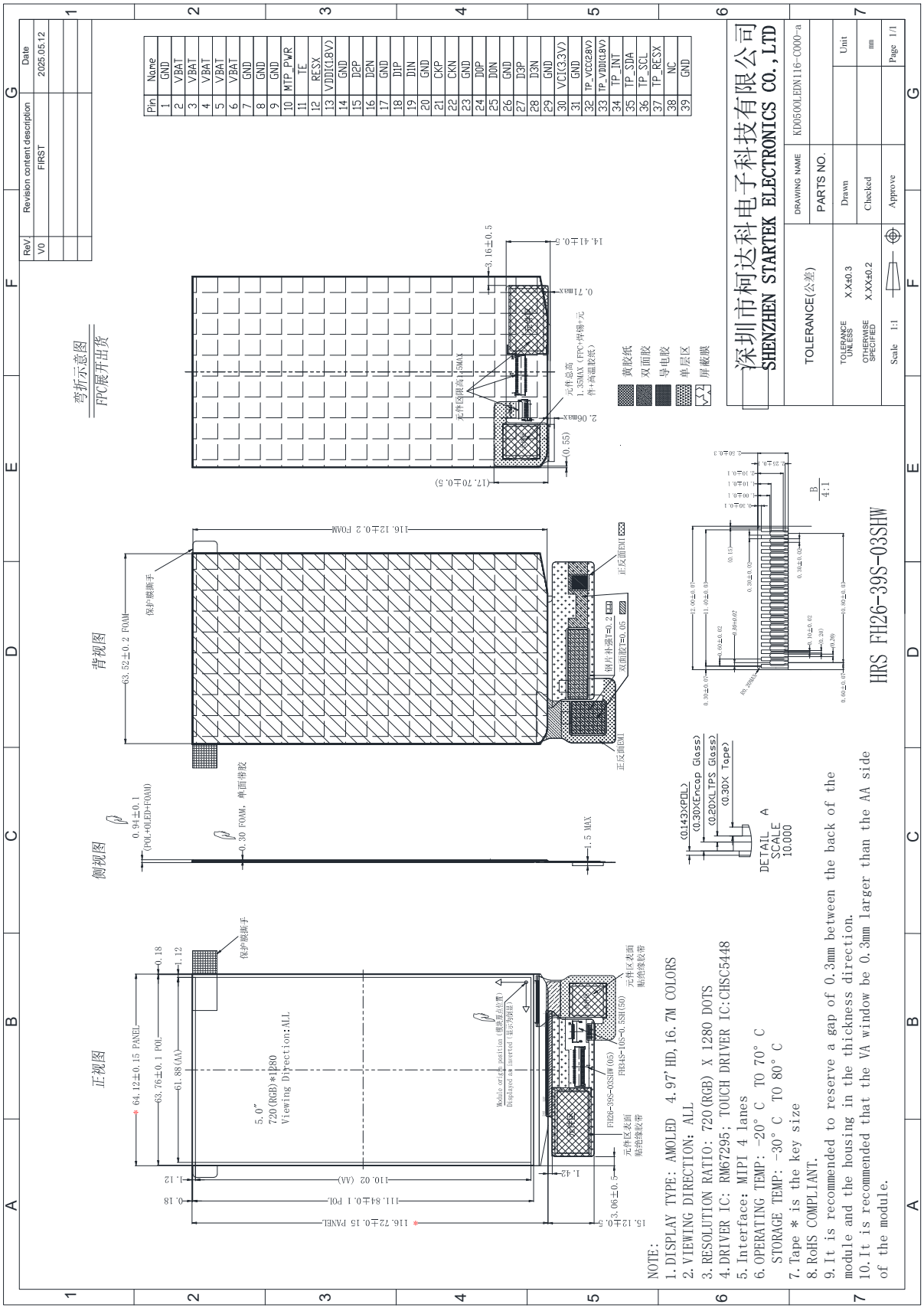
1. General Specifications

General Information Items	Specification	Unit	Note
	Main Panel		
Display area(AA)	61.884(H)*110.016(V) (5.0inch)	mm	
Technology Type	LTPS	-	
Display colors	16.7M (RGB x 8bits)	colors	
Number of pixels	720(RGB)*1280	dots	
Pixel arrangement	Rendering	-	
Pixel pitch	0.08595(H)*0.08595(V)	mm	
Viewing angle	ALL	o'clock	
Driver IC	RM67295(Raydium)	-	
LCM Interface	MIPI 4 lanes	-	
Touch IC	CHSC5448	-	
Touch Interface	IIC		
Touch Slave Address	0x2e(7bit)/0x5c(8bit)	-	
Touch screen	On-Cell		
Polarizer	Hard Coating Polarizer	-	
Glass thickness (LTPS/encapsulation glass)	0.20 / 0.30	mm	
Dimension outline	64.12*116.72*0.94	mm	
Encapsulation outline	64. 12 *113.82	mm	
Weight	14	g	
Operating temperature	-20～+70	℃	
Storage temperature	-30～+80	℃	

2. Block Diagram



3. Outline dimension



4. Input terminal Pin Assignment

Number	Signal	Description	I/O
1	GND	The power ground	P
2	VBAT	Power IC Input Voltage	P
3	VBAT	Power IC Input Voltage	P
4	VBAT	Power IC Input Voltage	P
5	VBAT	Power IC Input Voltage	P
6	VBAT	Power IC Input Voltage	P
7	GND	The power ground	P
8	GND	The power ground	P
9	GND	The power ground	P
10	MTP_PWR	Power supply for OTP. Leave the pin to open when not in use.	P
11	TE	Tear effect output	O
12	RESX	This signalwill reset thedevice and must be ap plied to properly initialize the chip. Active low.	I
13	VDDIO	Driver IC digital I/O supply	P
14	GND	The power ground	P
15	D2P	MIPI DSI data2+	I/O
16	D2N	MIPI DSI data2-	I/O
17	GND	The power ground	P
18	D1P	MIPI DSI data1+	I/O
19	D1N	MIPI DSI data1-	I/O
20	GND	The power ground	P
21	CLKP	MIPI DSI clock+	I
22	CLKN	MIPI DSI clock-	I
23	GND	The power ground	P
24	D0P	MIPI DSI data0+	I/O
25	D0N	MIPI DSI data0-	I/O
26	GND	The power ground	P
27	D3P	MIPI DSI data3+	I/O
28	D3N	MIPI DSI data3-	I/O
29	GND	The power ground	P
30	VCI	Driver IC analog supply	P
31	GND	The power ground	P

32	TP_VCC	TP IC digital power supply	P
33	TP_VDDI	TP IC digital I/O supply	P
34	TP_INT	Interrupt request to the host, orWakeup request from the host.	I/O
35	TP_SDA	I2C Data Input & Output	I/O
36	TP_SCL	I2C Clock Input	I/O
37	TP_RESX	External Reset,Low is Active	I
38	NC	NC	--
39	GND	The power ground	P

5. AMOLED Optical Characteristics

5.1 Optical specification

Item		Symbol	Conditions	Min.	Typ.	Max.	Unit	Remark
Brightness			Lv	450	500	--	cd/m^2	(5)
Brightness Uniformity			Lv	75	--	--	%	(5)
Contrast Ratio		CR	Normal to surface	60,000	--	--	--	(1)(2)
Color Filter Chromacicity	White	W_X	Normal to surface	-0.04	0.304	+0.04		CA-310 Test (1)(4)
		W_Y			0.323			
	Red	R_X			0.677			
		R_Y			0.323			
	Green	G_X			0.234			
		G_Y			0.683			
	Blue	B_X			0.140			
		B_Y			0.047			
Color Gamut			vs. NTSC	90	100	--	%	
Viewing angle			U/D/L/R CR≥ 1000	80	--	--	°	(1)(4)
Color shift (JNCD)			@ 45 degree	--	--	6		
Cross-talk				--	--	2	%	
Gamma			V(Gray)= 48,72,104, 132,164, 192,224,255	2.0	2.2	2.4		
Response time				--	--	2	ms	(1)(3)
Color Temperature				6537	7437	8337	K	
Flicker			Normal Θ = Φ = 0 °	--	--	-40	dB	
Image Retention				--	--	3	min	
Image Sticking				--	--	3	hrs	
Color Uniformity			Full White Δ u'v'	--	--	0.012		
Lifetime			Ta=25℃	30000	--	--	Hrs	

Measuring Condition

Measuring surrounding : dark room

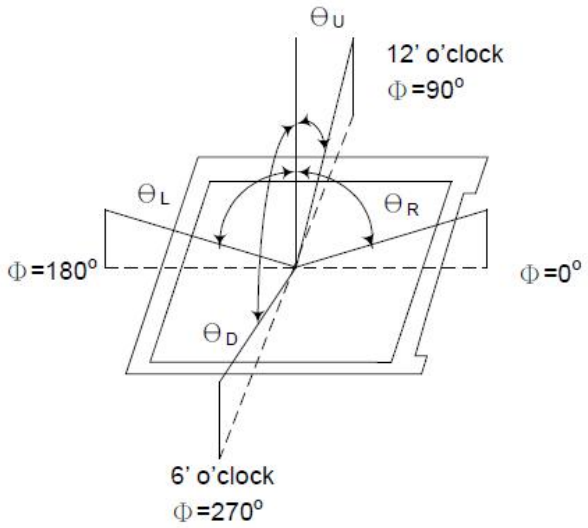
Ambient temperature : 25±2oC

15min. warm-up time.

Measuring Equipment

FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics.

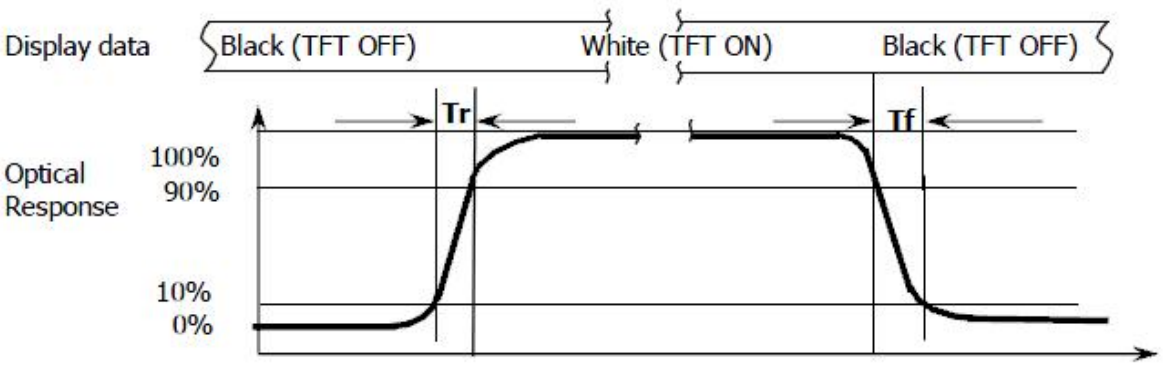
Note (1): Definition of Viewing Angle :



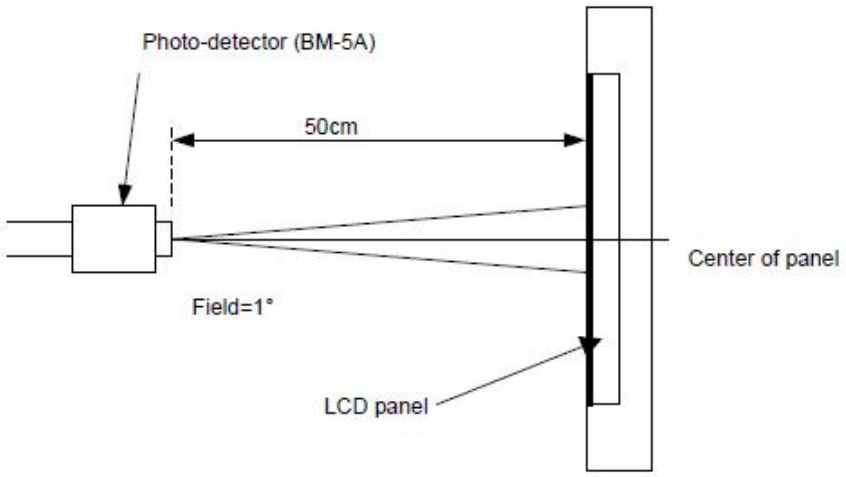
Note (2): Definition of Contrast Ratio(CR) :measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

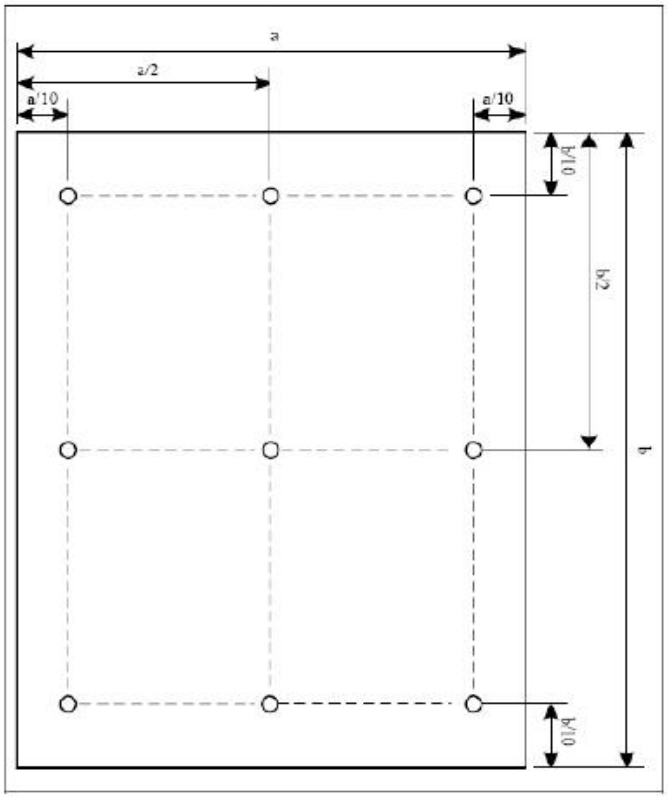
Note (3): Response Time



Note (4): Definition of optical measurement setup



Note (5)： Luminance Uniformity of these 9 points is defined as below:



$$\text{Uniformity} = \frac{\text{minimum luminance in 9 points (1-9)}}{\text{maximum luminance in 9 points (1-9)}}$$

$$\text{Luminance} = \frac{\text{Total Luminance of 9 points}}{9}$$

6. Electrical Characteristics

6.1 Absolute Maximum Rating

Maximum Ratings (Voltage Referenced to VSS) VSS=0V, Ta=25℃

Item	Symbol	Min.	Typ.	Max.	Unit	Note
Analog/boost power voltage	VCI	-0.3	--	5.5	V	Note1
I/O voltage	VDDIO	-0.3	--	5.5	V	
Power IC Input Voltage	VBAT	-0.3	--	6	V	
Operating temperature	Top	-20	--	70	℃	
Storage temperature	Tstg	-30	--	80	℃	

NOTE1: If the absolute maximum rating of even is one of the above parameters is exceeded even momentarily, the quality of the product may be degraded. Absolute maximum ratings, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the range of the absolute maximum ratings.

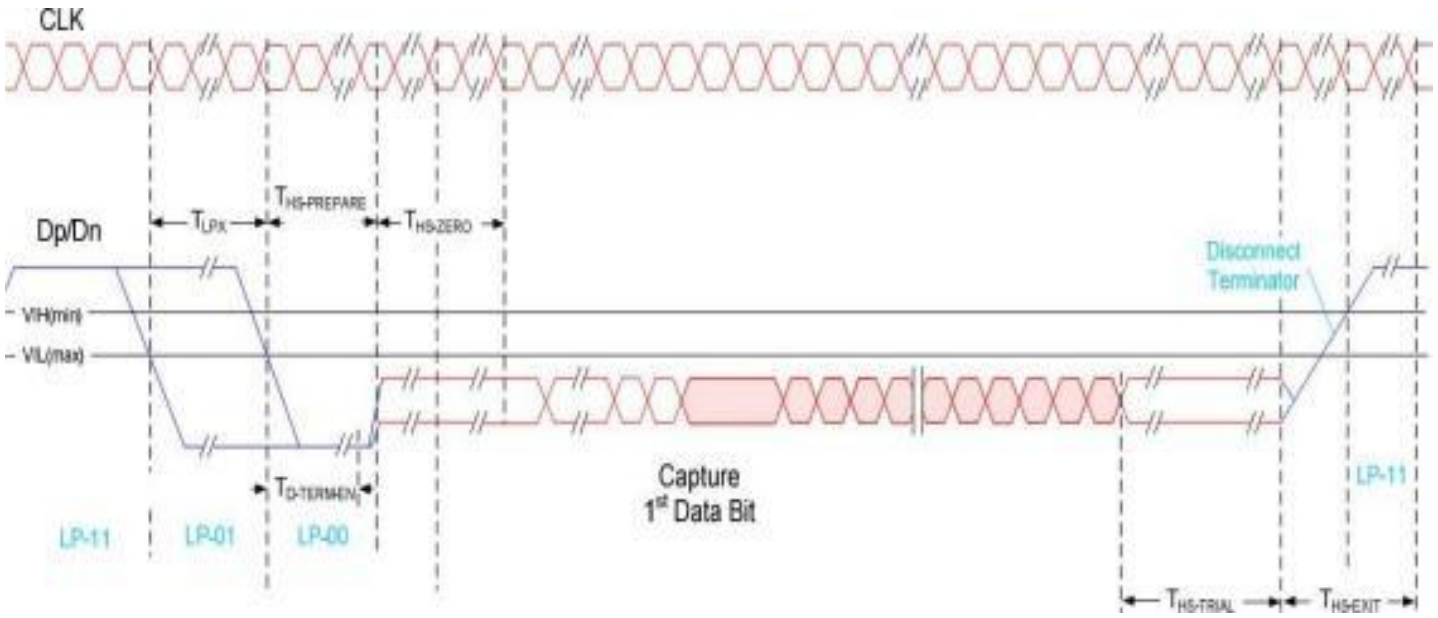
6.2 DC Electrical Characteristics

Item		Symbol	Min.	Typ.	Max.	Unit	Note
Power IC Input Voltage		VBAT	2.9	--	4.5	V	Ref
Digital Power supply		VDDIO	1.65	1.8	3.6	V	Ref
Analog Power supply		VCI	2.5	3.3	4.8	V	Ref
TP Power Supply voltage		TP_VCC	2.7	--	3.6	V	--
TP I/O Digital Voltage		TP_VDDI	1.65	--	3.6	V	
Full White @450 Nits	IVBAT=4V VCI=3.3V VDDIO=1.8V	IVBAT	--	300	380	mA	
		IVCI	--	2	3	mA	
		IVDDIO	--	15	20	mA	
TP Normal Operation	TP_VCC=3.3V TP_VDDI=1.8V	Iopr	--	16	--	mA	
TP Monitor		Imon	--	0.6	--	mA	
TP Sleep		Islp	--	40	--	uA	

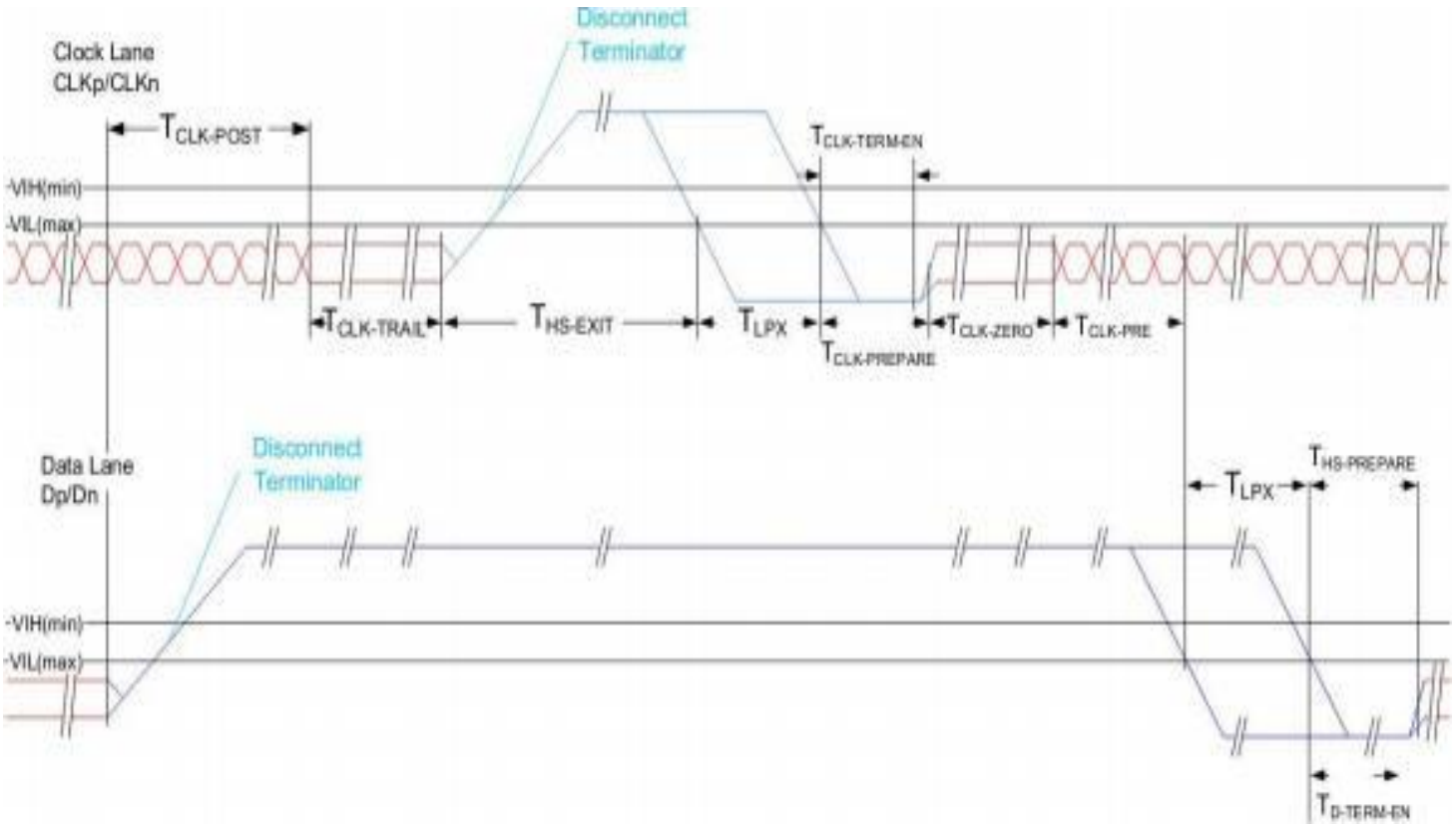
Note: The current and power consumption were tested under White pattern，25℃

7. AC Characteristics(MIPI)

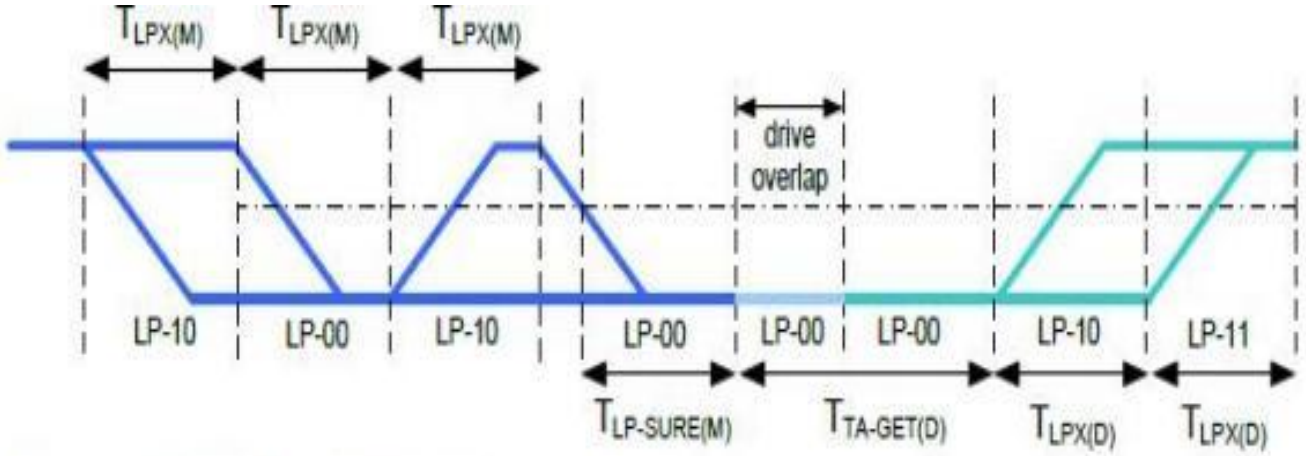
7.1 HS Data Transmission Burst



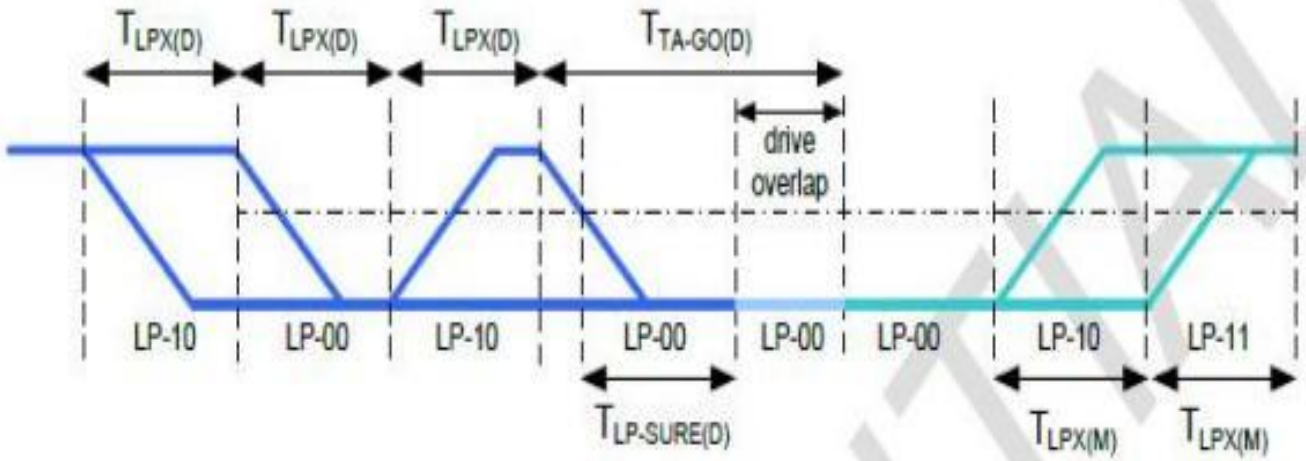
7.2 HS ClockTransmission



7.3 Turnaround Procedure



Bus turnaround (BAT) from MPU to display module timing



Bus turnaround (BAT) from display module to MPU timing

7.4 Timing Parameters

Timing Parameters:

Parameter	Description	Min	Typ	Max	Unit
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$.	60ns + 52*UI			ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the HS-0 state after the last payload clock bit of a HS transmission burst.	60			ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst.	300			ns
$T_{CLK-TERM-EN}$	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$		38	ns
$T_{CLK-PREPARE}$	Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.	38		95	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8			UI
$T_{CLK-PREPARE}$ + $T_{CLK-ZERO}$	$T_{CLK-PREPARE}$ + time that the transmitter drives the HS-0 state prior to starting the Clock.	300			ns
$T_{D-TERM-EN}$	Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$		35 ns +4*UI	
$T_{HS-PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission	40ns + 4*UI		85 ns + 6*UI	ns
$T_{HS-PREPARE}$ + $T_{HS-ZERO}$	$T_{HS-PREPARE}$ + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	145ns + 10*UI			ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst	60ns + 4*UI			ns

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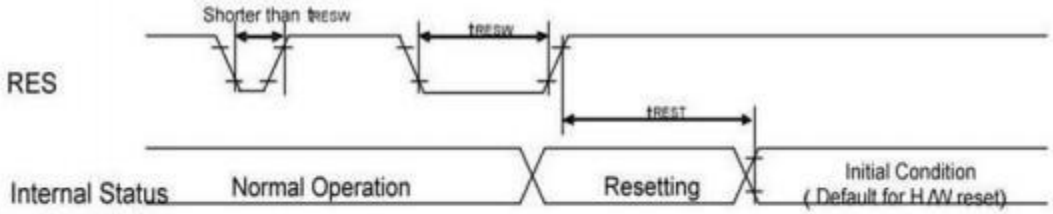
Low Power Mode :

Parameter	Description	Min	Typ	Max	Unit	Notes
$T_{LPX(M)}$	Transmitted length of any Low-Power state period of MCU to display module	50		**150**	ns	1,2
$T_{TA-SURE(M)}$	Time that the display module waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround.	$T_{LPX(M)}$		$2*T_{LPX(M)}$	ns	2
$T_{LPX(D)}$	Transmitted length of any Low-Power state period of display module to MCU	50		**150**	ns	1,2
$T_{TA-GET(D)}$	Time that the display module drives the Bridge state (LP-00) after accepting control during a Link Turnaround.		$5*T_{LPX(D)}$		ns	2
$T_{TA-GO(D)}$	Time that the display module drives the Bridge state (LP-00) before releasing control during a Link Turnaround.		$4*T_{LPX(D)}$		ns	2
$T_{TA-SURE(D)}$	Time that the MPU waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround.	$T_{LPX(D)}$		$2*T_{LPX(D)}$	ns	2

NOTE:

1. T_{LPX} is an internal state machine timing reference. Externally measured values may differ slightly from the specified values due to asymmetrical rise and fall times.
2. Transmitter-specific parameter

7.5 Timing requirements for RESETB



Reset input timing:

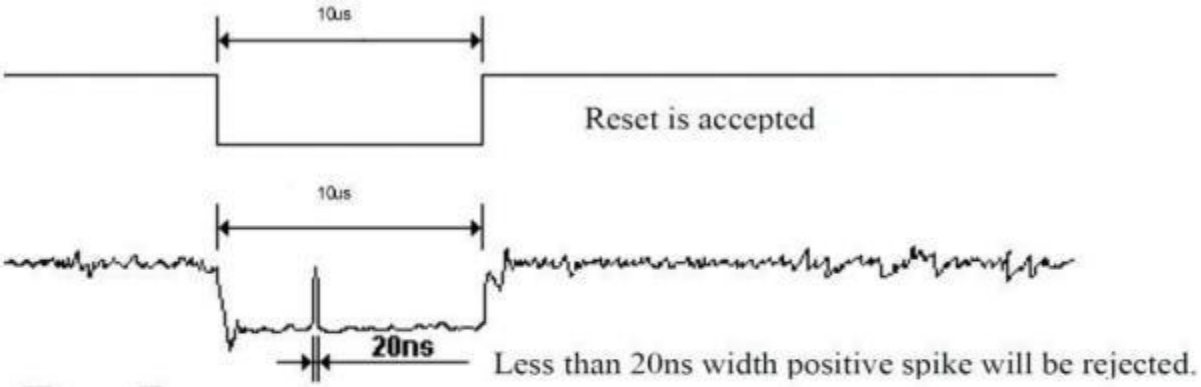
IOVCC=1.65 to 3.6V, VDD=2.5 to 3.6V, AGND=DGND=0V, Ta=-40 to 85℃

Symbol	Parameter	Related Pins	MIN	TYP	MAX	Note	Unit
t_{RESW}	*1) Reset low pulse width	RESX	10	-	-	-	µs
t_{REST}	*2) Reset complete time	-	-	-	5	When reset applied during Sleep in mode	ms
		-		-	120	When reset applied during Sleep out mode	ms

Note 1) Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below.

RESX Pulse	Action
Shorter than 5µs	Reset Rejected
Longer than 10µs	Reset
Between 5µs and 10µs	Reset starts (It depends on voltage and temperature condition.)

Note 2. During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then return to Default condition for H/W reset.
Note 3. During Reset Complete Time, data in OTP will be latched to internal register during this period. This loading is done every time when there is H/W reset complete time (tREST) within 5ms after a rising edge of RESX.
Note 4. Spike Rejection also applies during a valid reset pulse as shown below:

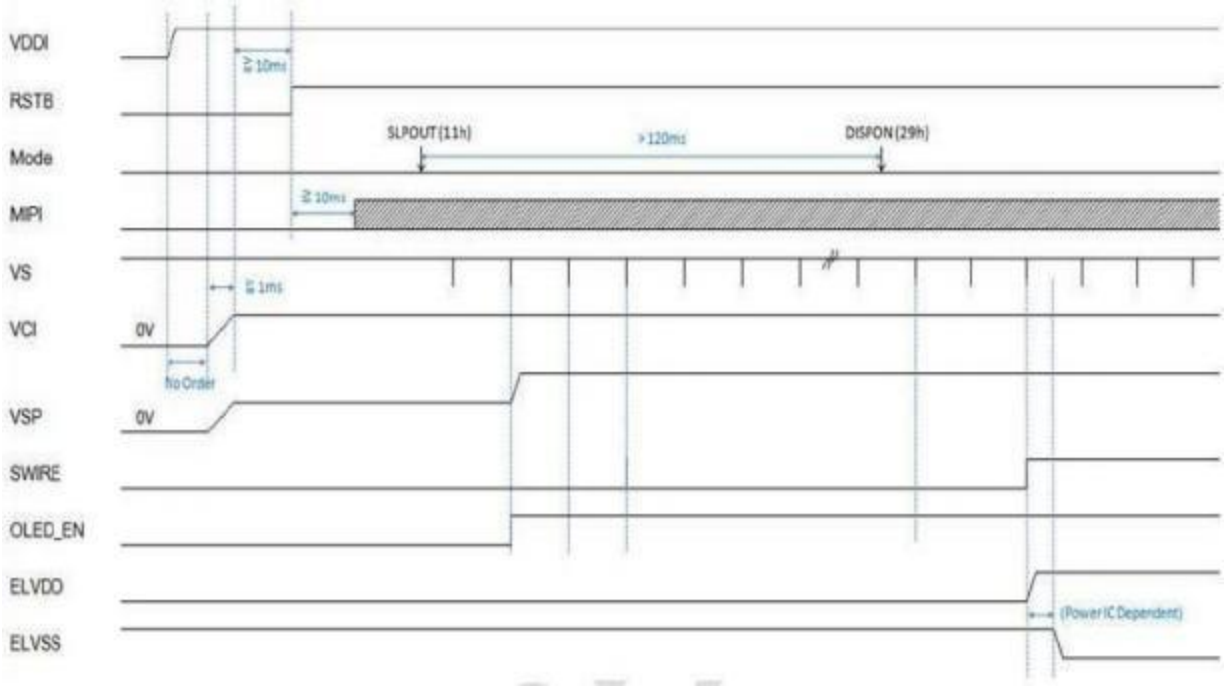


Note 5. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

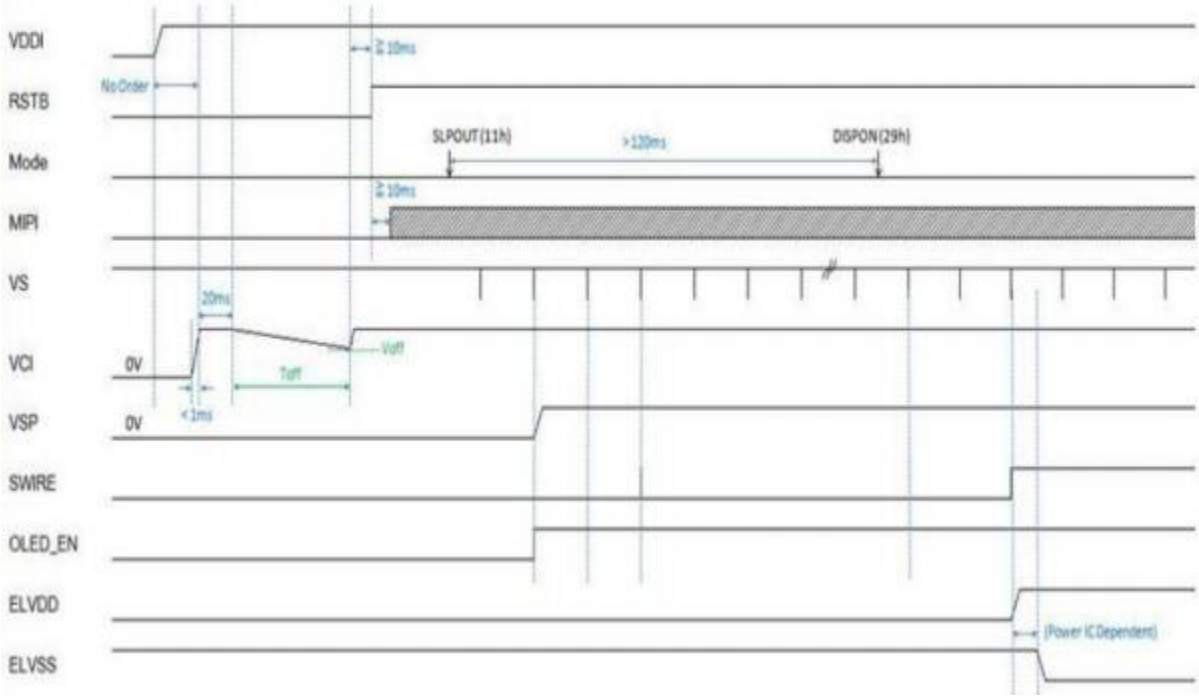
8. Recommended Operating Sequence

8.1 Power On Sequence

8.1.1 Power ON for Normal VCI Case: Tr> 1ms



8.1.2 Power ON for fast VCI Case: Tr< 1ms



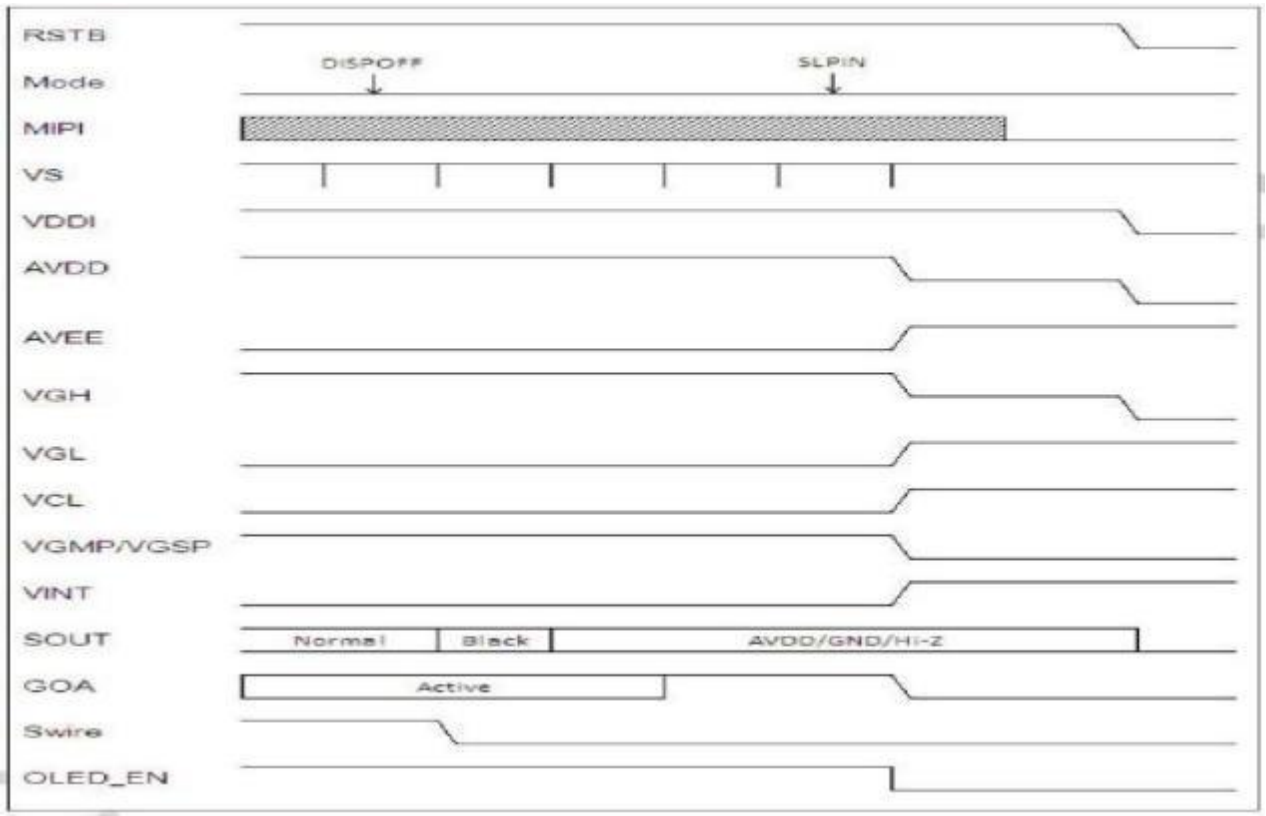
Note: If VCI's rising time is smaller than 1ms,it is suggested to use VCI H-L-H sequence to avoid the start-up issue due to fast VCI ramp up.

--the 1st H is suggested to be 20ms

--the L period is the time that can make Voff in between 2.6V and 1.5V.

(for example ,when the discharge time is 20ms and 200ms for VCI to drop below 2.6V and 1.5V, respectively. Then the Toff is suggested to be>20ms and <200ms.)

8.2 Power off Sequence



8.3 Brightness control

Inst/Para	R/W	Address		Date Type	Description
		MIPI	Other		
BRTCTRL	W	51h	5100h	Hex	Value form 0~255(FF)

9. AMOLED Module Out-Going Quality Level

9.1 VISUAL & FUNCTION INSPECTION STANDARD

9.1.1 Inspection conditions

Inspection performed under the following conditions is recommended.

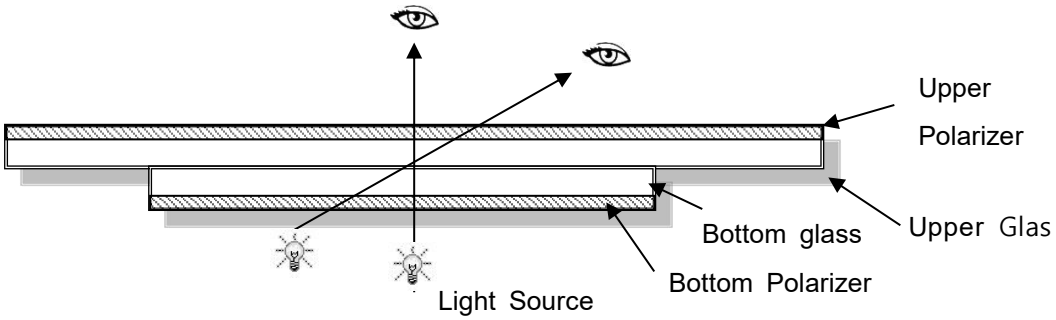
Temperature : 25±5℃

Humidity : 65%±10%RH

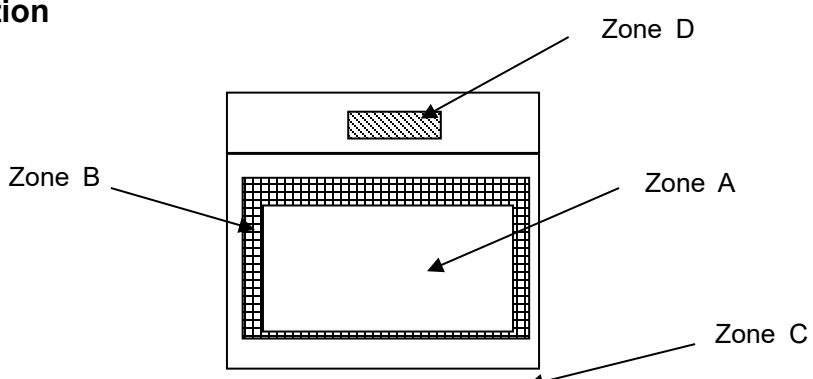
Viewing Angle : Normal viewing Angle.

Illumination: Single fluorescent lamp (300 to 700Lux)

Viewing distance:30-50cm



9.1.2 Definition



Zone A : Effective Viewing Area(Character or Digit can be seen)

Zone B : Viewing Area except Zone A

Zone C : Outside (Zone A+Zone B) which can not be seen after assembly by customer .)

Zone D : IC Bonding Area

Note:

As a general rule ,visual defects in Zone C can be ignored when it doesn't effect product function or ap pearance after assembly by customer

9.1.3 Sampling Plan

According to GB/T 2828-2012 ; , normal inspection, Class Ⅱ
AQL:

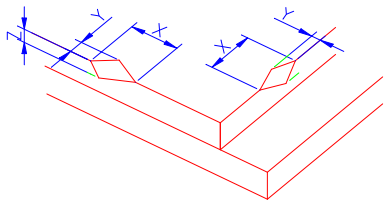
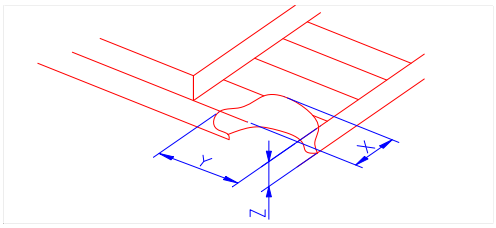
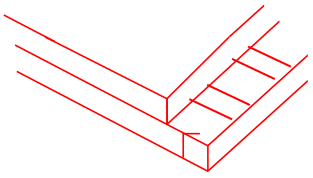
Major defect	Minor defect
0.65	1.5

No	Items to be inspected	Criteria	Classification of defects
1	Functional defects	1) No display, Open or miss line 2) Display abnormally, Short 3) Backlight no lighting, abnormal lighting. etc	Major
2	Missing	Missing components and etc	
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed,deformation and etc	
4	Color tone	Color unevenness, refer to limited sample	Minor
5	Spot/Line defect	Light dot,Dim spot,(Note1) Polarizer Air Bubble, Polarizer accidented spot and etc	
6	Soldering appearance	Good soldering , Peeling off is not allowed and etc	
7	AMOLED/Polarizer	Black/White spot/line, scratch, crack, etc.	

Note1: a) Light dot：Dots appear bright and unchanged in size in which AMOLED panel is displaying under black pattern.

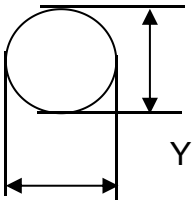
b) Dim dot: Dots appear dark and unchanged in size in which AMOLED panel is displaying under pure red, green, blue picture.

9.1.4 Criteria (Visual)

<table>
<tr><th>Number</th><th>Items</th><th>Criteria(mm)</th></tr>
<tr><td rowspan="3">1.0 AMOLED Crack/Broken
NOTE:
X: Length
Y: Width
Z: Height
L: Length of ITO,
T: Height of AMOELD</td><td>(1) The edge of AMOLED broken</td><td> 
<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td><Inner border line of the seal</td><td>≤T</td></tr></table></td></tr>
<tr><td>(2)AMOLED corner broken</td><td> 
<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td>≤L</td><td>≤T</td></tr></table></td></tr>
<tr><td>(3) AMOLED crack</td><td> 
Crack
Not allowed</td></tr>
</table>

2.0

Spot defect



Φ=(X+Y)/2

① light dot (black/white spot , pinhole, stain，etc.)

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.10	Ignore	Ignore	
0.10<Φ≤0.25	4(distance≧10mm)		
0.25<Φ≤0.35	3		
Φ>0.4	0		

② Dim spot (light leakage、dent、dark spot，etc)

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.1	Ignore	Ignore	
0.10<Φ≤0.25	4(distance≧10mm)		
0.25<Φ≤0.35	3		
Φ>0.4	0		

③ Polarizer accidented spot

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.2	Ignore		Ignore
0.3<Φ≤0.5	3(distance≧10mm)		
Φ>0.5	1		

④Polarizer Bubble

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.1	Ignore		Ignore
0.10<Φ≤0.25	2(distance≧10mm)		
0.25<Φ≤0.35	1		
Φ>0.4	0		

3.0 AMOLED Pixel defect

Pixel bad points

Item	Zone A	Acceptable Qt
Bright dot	Random	N≤2
	2 dots adjacent	N≤0
	3 dots adjacent	N≤0
Dark dot	Random	N≤3
	2 dots adjacent	N≤0
	3 dots adjacent	N≤0
Distance	1. Minimum Distance Between Bright dots. 2. Minimum Distance Between dark dots 3. Minimum Distance Between dark and bright dot.	5mm
Total bright and dark dot		N≤4

Note:

A) Bright dot：Dots appear bright and unchanged in size in which AMOLED panel is displaying under black pattern.

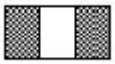
B) Dark dot: Dots appear dark and unchanged in size in which AMOLED panel is displaying under pure red, green, blue picture.

C) 2 dot adjacent = 1 pair = 2 dots

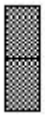
Picture:



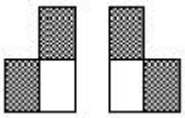
2 dot adjacent



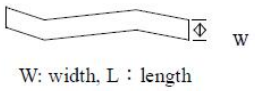
2 dot adjacent



2 dot adjacent (vertical)



2 dot adjacent (slant)

<table>
<tr>
<td>4.0</td>
<td>Line defect (AMOLED /Polarizer backlight black/white line, scratch, stain)
 

N：Count</td>
<td>
<table>
<tr><td rowspan="2">Width(mm)</td><td rowspan="2">Length(mm)</td><td colspan="3">Acceptable Qty</td></tr>
<tr><td>A</td><td>B</td><td>C</td></tr>
<tr><td>Φ≤0.05</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr>
<tr><td>0.05<W≤0.06</td><td>L≤5.0</td><td colspan="2">N≤3</td></tr>
<tr><td>0.07<W≤0.08</td><td>L≤4.0</td><td colspan="2">N≤2</td></tr>
<tr><td>W>0.08</td><td colspan="4">Define as spot defect</td></tr>
</table>
</td>
</tr>
<tr>
<td>5.0</td>
<td>Electronic Components SMT.</td>
<td>Not allow missing parts，solderless connection，cold solder joint，mismatch，The positive and negative polarity opposite</td>
</tr>
<tr>
<td>6.0</td>
<td>Display color& Brightness.</td>
<td>1. Color：Measuring the color coordinates, The measurement standard according to the datasheet or samples.
2. Brightness：Measuring the brightness of White screen, The measurement standard according to the datasheet or Samples.</td>
</tr>
</table>

Criteria (functional items)

Number	Items	Criteria (mm)
1	No display	Not allowed
2	Missing segment	Not allowed
3	Short	Not allowed
4	Backlight no lighting	Not allowed

10. Reliability Test Result

Item	Condition	Inspection after test
High Temperature Operating	70℃,96HR	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the AMOLED; 2.Non-display; 3.Missing segments/line; 4.Glass crack; 5.Current IDD is twice higher than initial value.
Low Temperature Operating	-20℃, 96HR	
High Temperature Storage	80℃, 96HR	
Low Temperature Storage	-30℃, 96HR	
High Temperature & High Humidity Operating	+60℃, 90% RH ,96 hours.	
Thermal Shock (Non-operation)	-10℃,30 min ↔ 60℃,30 min, Change time:5min 20CYC.	
ESD test	C=150pF, R=330,5points/panel Air:±8KV, 5times; Contact:±4KV, 5 times; (Environment: 15℃~35℃, 30%~60%).	
Vibration (Non-operation)	Frequency range:10~55Hz, Stroke:1.5mm Sweep:10Hz~55Hz~10Hz 2 hours for each direction of X.Y.Z. (6 hours for total) (Package condition).	
Box Drop Test	1 Corner 3 Edges 6 faces,80㎝(MEDIUM BOX)	

Remark:

1.The test samples should be applied to only one test item.

2.Sample size for each test item is 3~10pcs.

3.For Damp Proof Test, Pure water(Resistance > 10MΩ) should be used.

4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

5.Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

6. The color fading mura of polarizing filter should not care.

11. Cautions and Handling Precautions

11.1 Handling and Operating the Module

(1) When the module is assembled, it should be attached to the system firmly.
Do not warp or twist the module during assembly work.
(2) Protect the module from physical shock or any force. In addition to damage, this may cause improper operation or damage to the module and back-light unit.
(3) Note that polarizer is very fragile and could be easily damaged. Do not press or scratch the surface.
(4) Do not allow drops of water or chemicals to remain on the display surface.
If you have the droplets for a long time, staining and discoloration may occur.
(5) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.
(6) The desirable cleaners are water, IPA (Isopropyl Alcohol) or Hexane.
Do not use ketene type materials (ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanent damage to the polarizer due to chemical reaction.
(7) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth. In case of contact with hands, legs, or clothes, it must be washed away thoroughly with soap.
(8) Protect the module from static; it may cause damage to the CMOS ICs.
(9) Use finger-stalls with soft gloves in order to keep display clean during the incoming inspection and assembly process.
(10) Do not disassemble the module.
(11) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.
(12) Pins of I/F connector shall not be touched directly with bare hands.
(13) Do not connect, disconnect the module in the “Power ON” condition.

11.2 Storage and Transportation.

(1) Do not leave the panel in high temperature, and high humidity for a long time.
It is highly recommended to store the module with temperature from 0 to 35 ℃ and relative humidity of less than 70%
(2) Do not store the AMOLED module in direct sunlight.
(3) The module shall be stored in a dark place. When storing the modules for a long time, be sure to adopt effective measures for protecting the modules from strong ultraviolet radiation, sunlight, or fluorescent light.
(4) It is recommended that the modules should be stored under a condition where no condensation is allowed. Formation of dewdrops may cause an abnormal operation or a failure of the module.
In particular, the greatest possible care should be taken to prevent any module from being operated where condensation has occurred inside.

(5) This panel has its circuitry FPC on the bottom side and should be handled carefully in order not to be stressed.

12. Packing

----TBD----